

HW #4

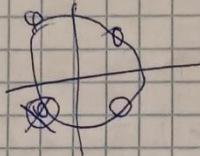
[1]  $-1-j$

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\cos \theta < 0$$

$$\sin \theta < 0$$

$$\cos \theta = \sin \theta$$



$$\theta = -\frac{3\pi}{4}$$

[3]  $j(1-j) = j - j^2 = j + 1$

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\cos \theta = \sin \theta > 0$$

$$\theta = \frac{\pi}{4}$$

[4]  $\sqrt{2} e^{j\frac{\pi}{4}}$

$$\sqrt{2} \left( \cos \frac{\pi}{4} + j \sin \frac{\pi}{4} \right) = \sqrt{2} \left( \frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}} \right) = 1 + j$$

[2]

$$(1-j)^2 = 1 - 2j + j^2 =$$

$$= -2j$$

$$r = \sqrt{4} = 2$$

$$\cos \theta = 0$$

$$\sin \theta < 0$$

$$\theta = -\frac{\pi}{2}$$

$$5) \underbrace{\{\sqrt{2} e^{j\frac{\pi}{4}}\}}_{j+1} - \{e^{-j(\frac{\pi}{2})}\}$$

from prev

$$e^{-j\frac{\pi}{2}} = e^{-j\frac{2\pi}{2}} = \cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2} = 0 - (-1) = 1$$

ans:  $j+1$

$$6) e^{j\frac{\pi}{4}} + e^{-j\frac{\pi}{4}}$$

$$\left( \cos \frac{\pi}{4} + j \sin \frac{\pi}{4} \right) + \left( \cos \frac{\pi}{4} - j \sin \frac{\pi}{4} \right) = 2 \cos \frac{\pi}{4} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

7)

$$y[-2] = (x[5-(-2)] + x[-2])u[-2-1] =$$

$$= 0 \text{ as } u[-3] = 0$$

8)

$$y[-1] = (\dots)u[-1-1] = 0$$

$$\text{as } u[-2] = 0$$

9)

$$y[0] = (\dots)u[0-1] = 0 \text{ as } u[-1] = 0$$

10)

$$y[1] = (x[4] + x[1])u[0] =$$

$$= 1 + 0 = 1$$

$$11) y[2] = (x[3] + x[2])u[1] =$$

$$= 2 + 2 = 4$$

$$12) y[3] = (x[2] + x[3])u[2] =$$

$$= 2 + 2 = 4$$

$$13) y[4] = (x[1] + x[4])u[3] =$$

$$= 0 + 1 = 1$$

$$14) y[5] = (x[0] + x[5])u[4] =$$

$$= 1 + 2 = 3$$

$$15) y[6] = (x[-1] + x[6])u[5] =$$

$$= 0 + 1 = 1$$





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| n      | -1   | 0    | 1    | 2    | 3    | 4    | 5 |
|--------|------|------|------|------|------|------|---|
| $x[n]$ |      | 1    | 2    | 0    | -1   |      |   |
| $h[n]$ | 2    | 0    | -2   |      |      |      |   |
|        |      | 2    | 0    | -2   |      |      |   |
|        |      |      | 4    | 0    | -4   |      |   |
|        |      |      |      |      | -2   | 0    | 2 |
| $y[n]$ | 0    | 2    | 4    | -2   | -6   | 0    | 2 |
|        | [28] | [29] | [30] | [31] | [32] | [33] |   |

[34] False